Database Design Process

Objectives

- **Database system development life cycle.**
- **Database Design stages**

Database system development life cycle.

- System definition
- Database design
- Database implementation
- Loading or data conversion
- Application conversion
- Testing and validation
- Operation
- Monitoring and maintenance

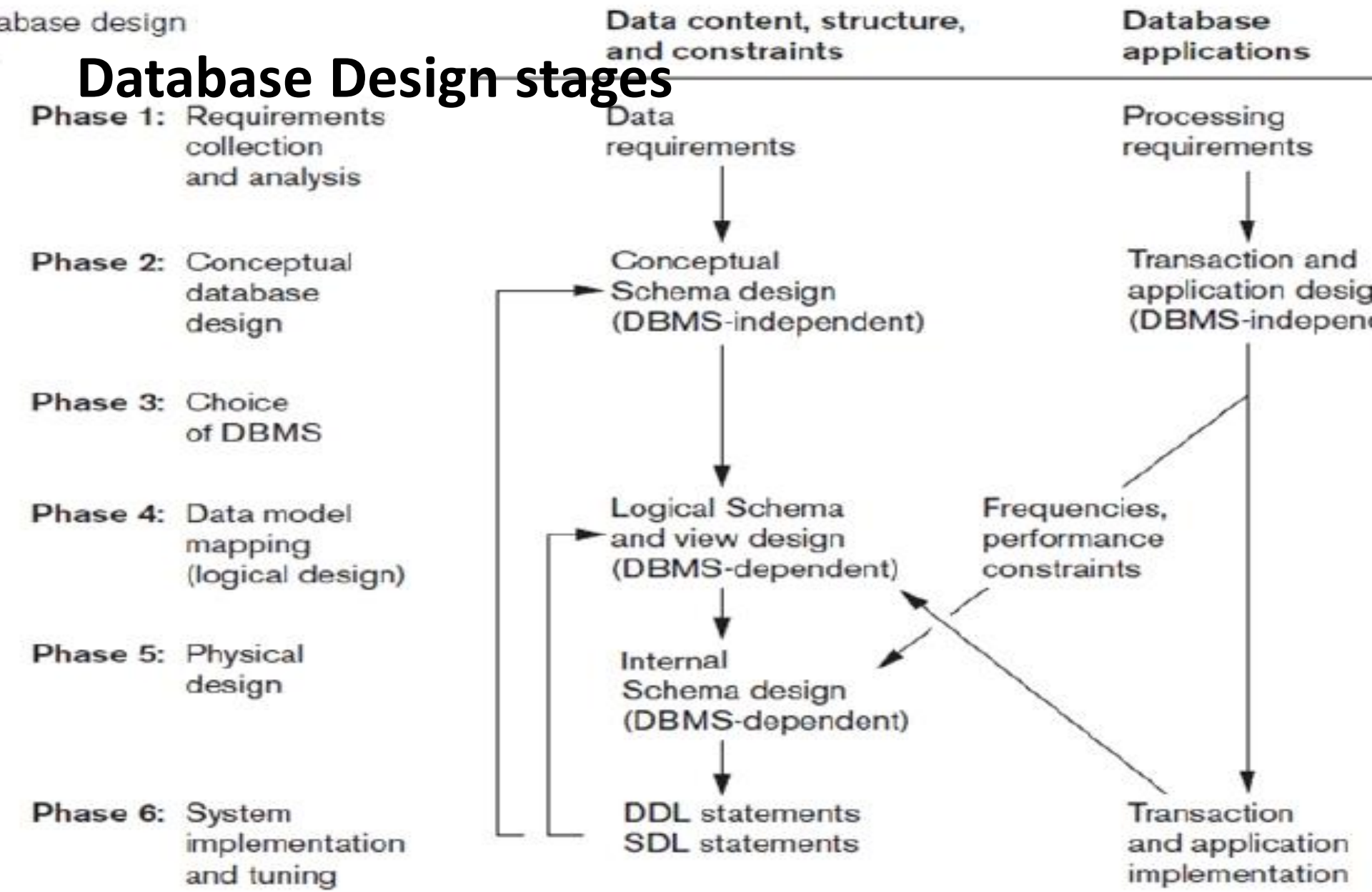
Database Design stages

- Design logical and physical structure of one or more databases
- Accommodate the information needs of the users in an organization for a defined set of applications
- Goals of database design
- Very hard to accomplish and measure
- Often begins with informal and incomplete requirements

Database Design stages

- Main phases of the overall database design
  - Requirements collection and analysis
  - Conceptual database design
  - Choice of a DBMS
  - Data model mapping (also called logical database design)
  - Physical database design

Database Design stages



Database Design stages

- Heart of the database design process
  - **Conceptual database design (Phase 2)**
  - **Data model mapping (Phase 4)**
  - **Physical database design (Phase 5)**
  - **Database system implementation and tuning (Phase 6)**

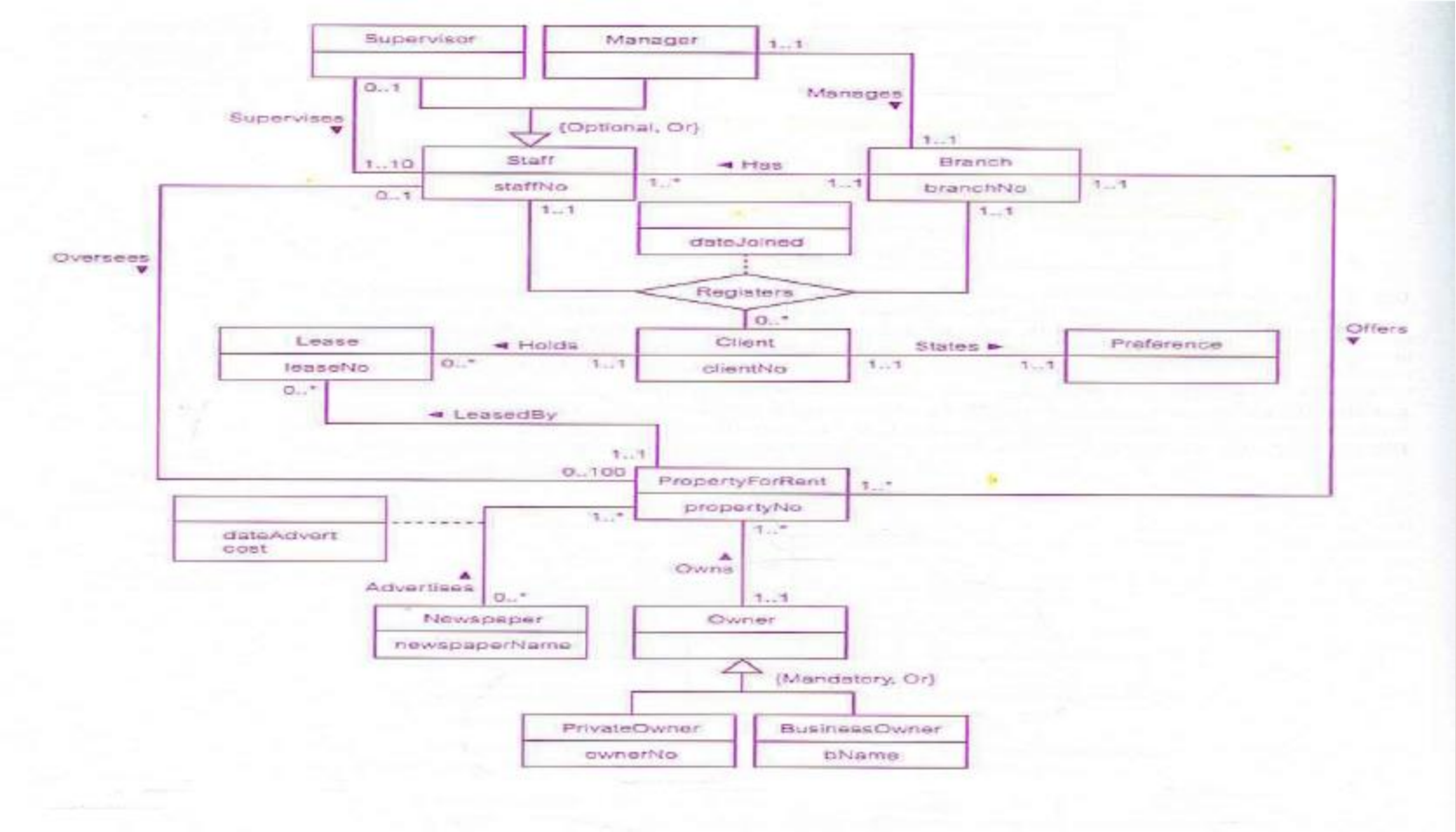
Phase 1: Requirements Collection and Analysis

- Identify application areas and user groups
- Study and analyze documentation
- Study current operating environment
- Collect written responses from users

Phase 2: Conceptual Database Design

- Conceptual Schema Design
- Developed a high level description of the data to be stored in the database along the constrains that are known to hold over this data.
- Transaction Design
  - Specify transactions at a conceptual level
  - Identify i **nput/output and functional behavior**
- Example ER diagram

Phase 2: Conceptual Database Design



Phase 3: Choice of a DBMS

- Costs to consider
- Software acquisition cost
- Maintenance cost
- Hardware acquisition cost
- Database creation and conversion cost
- Personnel cost
- Training cost
- Operating cost

Phase 4: Data Model Mapping (Logical Database Design)

- Convert the conceptual database design in to a database schema.
- In data model of selected DBMS.
- Stages
  - System-independent mapping

Phase 4: Data Model Mapping (LogicalDatabase Design)

Branch (branchNo, street, city, postcode, mgrStaffNo) **Primary Key** branchNo **Alternate Key** postcode **Foreign Key** mgrStaffNo **references** Manager(staffNo)	**Telephone** (telNo, branchNo) **Primary Key** telNo **Foreign Key** branchNo **references** Branch(branchNo)
Staff (staffNo, fName, lName, position, sex, DOB, salary, supervisorStaffNo, branchNo) **Primary Key** staffNo **Foreign Key** supervisorStaffNo **references** Staff(staffNo) **Foreign Key** branchNo **references** Branch(branchNo)	**Manager** (staffNo, mgrStartDate, bonus) **Primary Key** staffNo **Foreign Key** staffNo **references** Staff(staffNo)
PrivateOwner (ownerNo, fName, lName, address, telNo) **Primary Key** ownerNo	**BusinessOwner** (ownerNo, bName, bType, contactName, address, telNo) **Primary Key** ownerNo **Alternate Key** bName **Alternate Key** telNo
PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo, staffNo, branchNo) **Primary Key** propertyNo **Foreign Key** ownerNo **references** PrivateOwner(ownerNo) and BusinessOwner(ownerNo) **Foreign Key** staffNo **references** Staff(staffNo) **Foreign Key** branchNo **references** Branch(branchNo)	**Viewing** (c ientNo, propertyNo, dateView, comment) **Primary Key** clientNo, propertyNo **Foreign Key** clientNo **references** Client(clientNo) **Foreign Key** propertyNo **references** PropertyForRent(propertyNo)
Client (clientNo, fName, lName, telNo, prefType, maxRent) **Primary Key** clientNo	**Registration** (clientNo, branchNo, staffNo, dateJoined) **Primary Key** clientNo **Foreign Key** clientNo **references** Client(clientNo) **Foreign Key** branchNo **references** Branch(branchNo) **Foreign Key** staffNo **references** Staff(staffNo)
Lease (leaseNo, paymentMethod, depositPaid, rentStart, rentFinish, cl entNo, propertyNo) **Primary Key** easeNo **Alternate Key** propertyNo, rentStart **Alternate Key** c ientNo, rentStart **Foreign Key** clientNo **references** Client(clientNo) **Foreign Key** propertyNo **references** PropertyForRent(propertyNo) **Derived** depos t (PropertyForRent.rent*2) **Derived** duration (rentFinish – rentStart)	**Newspaper** (newspaperName, address, telNo, contactName) **Primary Key** newspaperName **Alternate Key** telNo
Advert (propertyNo, newspaperName, dateAdvert, cost) **Primary Key** propertyNo, newspaperName, dateAdvert **Foreign Key** propertyNo **references** PropertyForRent(propertyNo) **Foreign Key** newspaperName **references** Newspaper(newspaperName)	

Phase 5: Physical Database Design

- Choose specific file storage structures and access paths for the database files.
- Achieve good performance.

Phase 6: Database System Implementation and Tuning

- Typically responsibility of the DBA
- Compose DDL
- Load database
- Convert data from earlier systems
- Database programs implemented by application programmers
- Most systems include monitoring utility to collect performance statistics

Summary

- Database design life cycle
- Database design process
  - Requirement analysis
  - Conceptual database design
  - Logical database design
  - Physical database design
  - Security design

Questions

- Briefly explain database development life cycle process?
- What are the main steps in database design phase?